RAGA VINAY DEWARASETTY

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Summary

Engineering student with practical experience in building small embedded and AI-based projects. I have worked with ESP32, STM32, and Arduino boards to make real-time systems like home security and automation. I've also used Python and OpenCV for basic face detection and bot integration. Comfortable with sensors, interrupts, simple firmware coding, and basic AI model usage. I enjoy learning new technologies and improving my skills through hands-on work.

Education

B. Tech. in Electronics and Communication Engineering 2022 – 2026 | Bengaluru, India
Amrita Vishwa Vidyapeetham, Bengaluru
Current CGPA: 7.15
Minor degree: Artificial Intelligence Machine Learning (2023-2026)

Secondary School Education 2020 – 2022 | Hyderabad, India
Sri Chaitanya Educational Institutions, Hyderabad
95.1%

Skills

Embedded Systems — Microcontroller (ESP32,Arduino-UNO, NANO) | UART, I2C Communication | Sensor Interfacing
Simulation & EDA Tools — LTSpice | Keil MicroVision | Proteus | MATLAB | STM32CubeIDE | Arduino IDE
Programming — Python | Embedded C | C++

Projects

Smart Solar EV Charging with Dual Battery Switching & Solar Tracking (ESP32) 02/2026 – 05/2026

- Designed an automated dual-battery switching system with threshold-based logic and real-time performance monitoring, validated through structured testing against defined specs.
- Built an IoT web dashboard (JSON/HTTP) for live tracking of system KPIs — voltage, charge state, status — enabling remote monitoring and troubleshooting.
- Integrated the battery switching, monitoring, and communication modules into a unified prototype and verified their operation under different operating conditions.

Wireless Charging System for Electric Vehicles (WPT-Based) 08/2025 – 11/2025

- Simulated a wireless EV charging system using MATLAB/Simulink to study power transfer efficiency and coil alignment effects.
- Built a basic hardware prototype using transmitter–receiver coils, IR-based detection, and Arduino-controlled relay switching.
- Verified system operation using LCD feedback and practical testing of charging behavior.

Smart Home Security System using ESP32-CAM 02/2025 – 04/2025

- •Developed a project using Arduino IDE interfacing Arduino UNO and ESP32, implemented ISR-based ultrasonic triggering, configured WiFi-based real-time image streaming, and integrated servo control using PWM•Implemented real-time face detection and automated Telegram bot using Python and OpenCV.•Designed for modularity and scalability with future support for face recognition and cloud integration.

Certifications

- Apna College: Data Structures and Algorithms (C++)
- Udemy: The Full Stack Web Development Bootcamp
- Udemy: 100 days Python Bootcamp